

THE EFFECTS OF EXERCISE SELECTION AND REST INTERVAL ON POSTACTIVATION POTENTIATION OF VERTICAL JUMP PERFORMANCE

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ABSTRACT

McCann, MR and Flanagan, SP. The effects of exercise selection and rest interval on postactivation potentiation of vertical jump performance. *J Strength Cond Res* 24(5): 1285–1291, 2010—The purpose of this study was to determine if a power exercise would lead to greater postactivation potentiation (PAP) than a strength exercise, if a 4- or 5-minute rest interval led to greater PAP, the extent to which PAP was an individual phenomenon, and the effect of PAP on the ground reaction force (GRF) during a vertical jump (VJ). Subjects included 16 volleyball athletes (8 men and 8 women) from a Division I university. Participants were instructed to complete a pre-exercise countermovement jump for height. After the VJ, subjects performed 5 repetitions of either the back squat or hang clean (midthigh) with a load equal to their 5 repetition maximum (5RM). After the 5RM resistance exercise, countermovement jumps were completed at 4 or 5 minutes after the back squat or hang clean. Ground reaction force was measured using a force platform embedded in the ground, whereas VJ height was assessed using a Vertec jump apparatus. Data were analyzed using a factorial analysis of variance with repeated measures. Results revealed that there was no consistent rest interval or exercise that produced the largest increase in VJ height for all subjects, and there were no apparent differences because of gender. The condition that produced the largest increase in VJ height for each individual resulted in an average increase of 5.7% (2.72 ± 1.21 cm; $p < 0.001$). There was no significant difference ($p > 0.05$) in peak GRF, and no significant correlation ($r = -0.110$, $p = 0.707$) between the increase in VJ height and increase in peak GRF. Results suggest that individually determining complex training variables will increase VJ height, thus acutely enhancing athletic performance.

KEY WORDS complex training, muscle potentiation, countermovement jump, back squat, hang clean

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INTRODUCTION

Complex training is the combination of a heavy loaded resistance exercise followed by a biomechanically similar plyometric exercise, and is a training method used to improve athletic performance in sports requiring explosive vertical movements, such as volleyball and basketball. It is suggested that after a heavy loaded resistance exercise (i.e., 3–5 repetition maximum [RM]), there is a postactivation potentiation (PAP) effect, which increases the amount of force produced and height during a vertical jump (VJ). Postactivation potentiation is the change in force–time and force–velocity characteristics of skeletal muscle, which leads to increased power output. However, to date, the evidence for PAP leading to improved motor performance remains equivocal (for extensive reviews, see [11,19]). There may be at least 4 reasons for these findings.

First, complex training involves the pairing of a heavy resistance exercise with a high power exercise. Because of the nature of the force–velocity relation during concentric muscle actions (10), heavy resistance exercise will involve a high force with a low movement velocity, whereas movements such as vertical jumping require a (relatively) low force with a high movement velocity. This disparity may not create the conditions optimal for PAP.

First, the type of exercise has been shown to affect the PAP. Traditionally, complex training involves the pairing of a heavy resistance exercise with a high power exercise. Because of the nature of the force–velocity relation during concentric muscle actions (10), heavy resistance exercise will involve a high force with a low movement velocity, whereas movements such as vertical jumping require a (relatively) low force with a high movement velocity. This disparity may not create the conditions optimal for PAP. Although isometric muscle actions led to a greater PAP than dynamic muscle actions (16), possibly because of the higher force generation, to our knowledge the effect of an explosive exercise on PAP has not been investigated. Therefore, the effect of an exercise such as the hang clean on PAP needs to be explored.

Second, PAP exists concurrently with fatigue (2). Theoretically, both PAP and fatigue would be maximal immediately after heavy resistance exercise, with fatigue having a higher rate of abatement. Motor performance would be

maximized when the difference between PAP and fatigue is greatest (i.e., PAP effects are greater than fatigue effects), which may require a rather precise rest interval.

Thus far, the establishment of an optimal rest interval for PAP remains elusive. Investigations have examined rest intervals as short as 10 seconds (12) and as long as 20 minutes (13). Some studies have found significant differences in jump height postexercise at 4 minutes of rest (20), whereas others have not (12). Similarly, 18.5 minutes postactivation has led to significant increases (3), whereas 20 minutes postactivation has not (13). These disparate results may be because of differences in methodology and the subjects tested.

Third, the PAP phenomenon may be highly individualized (3,17). Some individuals may have a greater PAP effect than fatigue effect, whereas others may have a greater fatigue effect than PAP effect for the same rest interval. The differences would wash out and show no effect. Therefore, it may be necessary to examine subjects as individuals (1) to determine which combination of exercise and rest interval would maximize motor performance for each person.

For example, Chiu et al. (3) found that PAP was larger for athletes than for recreationally trained individuals. Post-activation potentiation appears to be larger in individuals (9) and muscles (15) with a preponderance of type II muscle fibers. Gender does not appear to affect PAP (16), but few studies have examined this subject characteristic.

Finally, the link between muscle activation and performance is the ground reaction force (GRF). It is this interaction with the support that ultimately determines performance. Therefore, examining the GRF is crucial when determining the underlying mechanisms of how PAP influences motor performance. Most studies have only examined a single parameter of the GRF profile, namely, the peak GRF. Garhammer and Gregor (6) have shown that peak GRF is a poor indicator of VJ height. It is possible that PAP may influence performance in ways other than simply increasing peak GRF, and the entire shape of the GRF-time curve needs to be examined.

The purpose of this investigation was to address these 4 areas. We wished to examine the effect of exercise selection (strength vs. power), rest intervals, and gender on improvements in jump height. The intent was not to determine the optimal conditions for PAP, but to examine group mean and individual responses to exercise selection and rest interval. Therefore, both group mean and single-subject data were compared, and the GRF curves were analyzed in an attempt to explain the results.

METHODS

Experimental Approach to the Problem

Subjects completed a protocol with 4 conditions: 2 exercises (squat, hang clean) with 2 rest periods (4 minutes, 5 minutes). Before and after each condition,

subjects completed a countermovement VJ for height while standing on a force platform. VJ height and GRF characteristics were compared between pre and post for all 4 conditions using a combination of group mean and single-subject analyses.

Subjects

Subjects included 16 NCAA Division I volleyball athletes (8 women; 8 men) over the age of 18, whose characteristics are presented in Table 1. All subjects were out of season and had at least 1 year of previous experience with resistance and plyometric training, which included both the squat and clean exercise. Subjects did not possess any lower extremity injury. Because of injuries sustained outside of the study, 1 male and 1 female subject were unable to complete the hang clean portion of testing. Therefore, results are presented on 14 subjects. Subjects did not complete a training session 48–72 hours before testing. Permission from the head strength and conditioning coach and a signed informed consent form from the athletes were obtained before testing commenced. The informed consent form notified the athletes of the testing procedures, which was approved by this institution's Standing Advisory Committee on the Protection of Human Subjects, and the risks and benefits involved with their participation. The strength and conditioning coach informed the volleyball coaches of their athletes' participation.

Instrumentation

The VJs were performed on a Kistler Force Platform (Amherst, NY, USA) embedded in the floor. Ground reaction forces were collected and saved on a desktop computer where data were analyzed using Datapac (Run Technologies, Laguna Hills, CA, USA). All data were collected at 1,200 Hz. Vertical jump height was determined by using a Vertec jump apparatus.

Procedures

Subjects were randomized and assigned to 1 of 4 groups. The first 2 groups (A and B) completed a back squat on day 1 and a hang clean on day 2. The second 2 groups (C and D) completed a hang clean on day 1 and a back squat on day 2. The first and third groups (A and C) had a rest interval of 4 minutes, followed by 5 minutes on the first day. The second and fourth groups (B and D) had a rest interval of 5 minutes, followed by 4 minutes on the first day. The exercise and order of the rest intervals were reversed on the second day. There was

TABLE 1. Mean $\pm$ SD of participant physical characteristics.

Gender	Age (y)	Height (cm)	Mass (kg)
Women	19.14 $\pm$ 0.38	177.44 $\pm$ 7.10	72.27 $\pm$ 9.41
Men	20.86 $\pm$ 1.77	195.22 $\pm$ 5.37	89.42 $\pm$ 8.40

TABLE 2. Randomization of exercises and rest intervals by group.

	Group			
	A	B	C	D
Day 1				
Exercise	Squat	Squat	Clean	Clean
First rest period (min)	4	5	4	5
Second rest period (min)	5	4	5	4
Day 2				
Exercise	Clean	Clean	Squat	Squat
First rest period (min)	5	4	5	4
Second rest period (min)	4	5	4	5

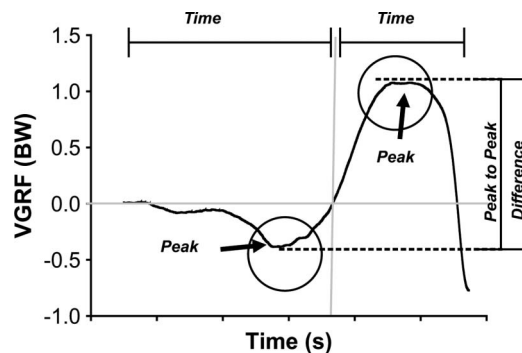


Figure 1. Characteristics of the vertical ground reaction force studied in this investigation.

TABLE 3. Mean $\pm$ SD of vertical jump height (cm) for the group ($n = 14$).

Rest interval (min)	Vertical jump height (cm)	
	Pre	Post
4	52.84 $\pm$ 12.80	53.98* $\pm$ 11.63
5	52.48 $\pm$ 12.48	53.16 $\pm$ 12.78

*Significant difference between prejump and post-jump values ($p < 0.05$).

at least 48 hours between testing sessions. Testing was held in the university's weight room and biomechanics laboratory.

On day 1, subjects in group A completed a dynamic warm-up that consisted of exercises beginning with lighter resistance movements followed by higher resistance, higher

velocity movements. Back squats were completed at 50% 5RM for 5 repetitions, and 80% 5RM for 5 repetitions. The 5RM represents the maximal load that the subject could successfully complete for 5 repetitions with correct form. Each subject's 5RM was determined from a familiarization session using a protocol adapted from Stone and O'Bryant (18). Subjects then completed 2 sets of 5 VJs. After the warm-up was complete, subjects had 5 minutes of rest.

Subjects then performed a VJ for height. After the VJ, subjects performed 5 repetitions of the back squat with a load equal to their 5RM. After the 5RM resistance exercise, VJs were completed 4 minutes after the back squat. On completion of the first part of testing, subjects had 15 minutes of rest before completing the same procedures as before, using a different intracomplex rest interval (5 minutes). A cooldown consisting of static stretching was administered after completion of the testing procedures.

On day 2, subjects in group A performed the same warm-up and testing procedures as on day 1, using the hang clean. At least 48 hours of rest was taken between testing sessions. A

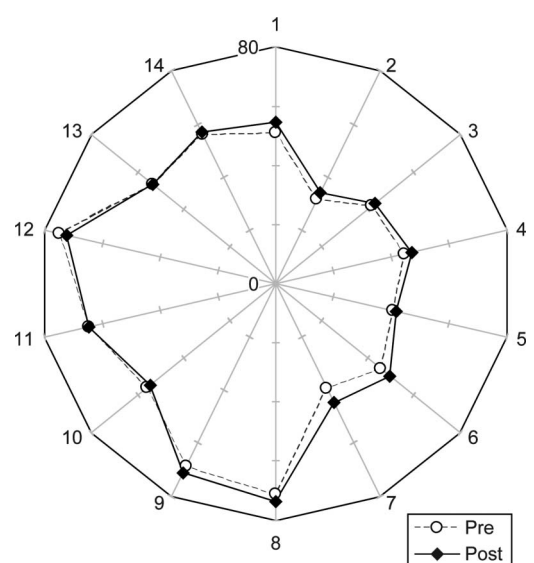


Figure 2. Radar plot for the 4-minute squat condition. The angular spokes indicate the individual subjects (1–7 women and 8–14 men). The radius (distance from center to plot) indicates vertical jump height (in centimeters). Open circles indicate pre-exercise jump height. Closed diamonds indicate postexercise jump height.

similar protocol was used for the other groups, with the exercise and rest intervals varying according to group assignment (Table 2). A certified strength and conditioning specialist supervised the subjects during the testing procedures to ensure a safe environment and limit the risk of potential injuries.

Statistical Analyses

First, intraclass correlation coefficients (ICCs) were calculated for the 4 pre-exercise jump heights. Vertical jump heights were then analyzed using a 2×3 (exercise selection $\times$ rest intervals) factorial analysis of variance (ANOVA) with repeated measures ($p \leq 0.05$). To eliminate any washout effects and analyze the effect of gender, the condition (exercise and rest interval) that produced the largest increase in VJ height was analyzed pre and postexercise. Vertical jump heights were compared using a 2×2 (gender $\times$ time) factorial ANOVA with repeated measures ($p \leq 0.05$). Chi-square analyses (gender $\times$ exercise; gender $\times$ rest interval) were used to if gender had an effect on the particular condition that produced the largest increase in VJ.

Next, we wished to determine how improvements in jump height (if present) during the condition that led to greatest improvements in VJ height were brought about by changes in the VGRF, and if these effects were influenced by gender. The VGRF was first partitioned into 2 phases—an unweighting (UW) (where the VGRF was negative) and a weighting (W) phase (where the VGRF was positive). For each phase, the duration (time) of the phase, peak amplitude, and impulse of

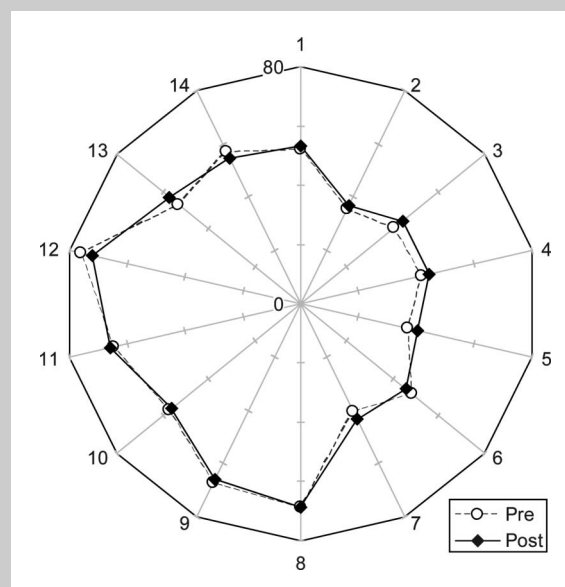


Figure 4. Radar plot for the 4-minute power clean condition. The angular spokes indicate the individual subjects (1–7 women and 8–14 men). The radius (distance from center to plot) indicates vertical jump height (in centimeters). Open circles indicate pre-exercise jump height. Closed diamonds indicate postexercise jump height.

the VGRF were determined. Additionally, the differences in the peak amplitudes between phases were determined by subtracting the negative peak from the positive peak. These characteristics are represented in Figure 1 and were

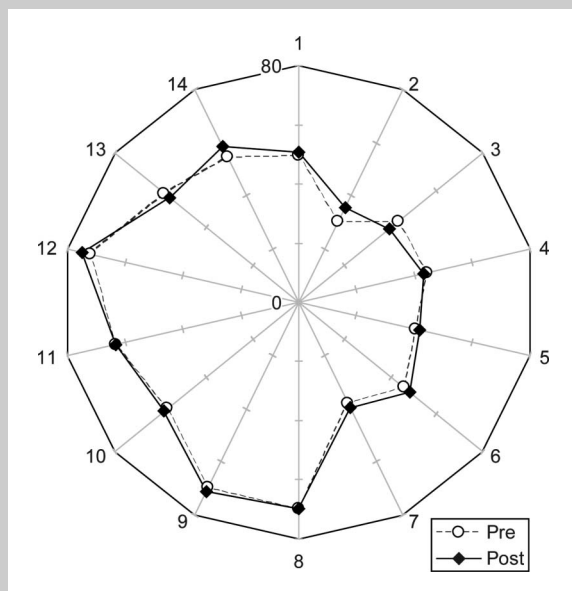


Figure 3. Radar plot for the 5-minute squat condition. The angular spokes indicate the individual subjects (1–7 women and 8–14 men). The radius (distance from center to plot) indicates vertical jump height (in centimeters). Open circles indicate pre-exercise jump height. Closed diamonds indicate postexercise jump height.

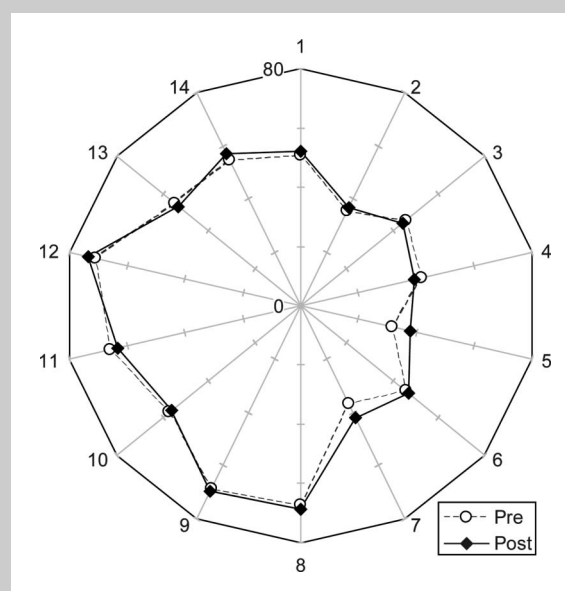


Figure 5. Radar plot for the 5-minute power clean condition. The angular spokes indicate the individual subjects (1–7 women and 8–14 men). The radius (distance from center to plot) indicates vertical jump height (in centimeters). Open circles indicate pre-exercise jump height. Closed diamonds indicate post-exercise jump height.

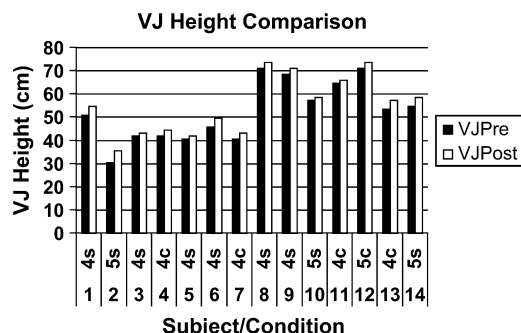


Figure 6. Individual comparison of vertical jump height (pre vs. post) using the condition that produced the largest increase in jump performance. 4s: 4-minute back squat; 4c: 4-minute clean; 5s: 5-minute back squat; and 5c: 5-minute clean. Subjects 1–7 were women and subjects 8–14 were men.

compared between the pre and postexercise conditions using a 2×2 (gender $\times$ time) factorial ANOVA with repeated measures ($p \leq 0.05$). All analysis was conducted using SPSS (Chicago, IL).

RESULTS

Intraclass correlation coefficients for the pre-exercise jump height values were high (average measures ICC = 0.994; $p < 0.001$).

Vertical Jump Height

For VJ height (Table 3), there was a significant main effect for rest interval ($p = 0.040$), but no significant main effect for exercise ($p = 0.694$) and no significant interaction ($p = 0.475$). Post hoc analysis revealed that the 4-minute rest interval was significantly different from prejump data ($p = 0.016$), whereas the 5-minute rest interval had no significant difference ($p = 0.117$) on VJ height.

The results for each condition were highly individualistic. The data for each condition are presented in the radar plots in Figures 2–5. No consistent exercise or rest interval produced the largest increase in VJ height (Figure 6) for all subjects. However, when the condition (exercise and rest interval) with the largest increase in VJ height was selected, there was a 5.7% (2.72 ± 1.21 cm) increase in the group mean data ($p < 0.001$).

Ground Reaction Force Data

When comparing the pre- and post-GRF characteristics (Table 4) for the condition that produced the largest increase in VJ height, there were no statistically significant differences in the time of the UW phase ($p = 0.517$), but the time of the W phase decreased by 6% ($p = 0.043$). There was a 17.2% increase in the peak amplitude of the VGRF during the UW phase ($p = 0.003$), but no statistically significant differences in the peak VGRF during the W phase ($p = 0.159$). This resulted in a 5.25% increase in the peak-to-peak differences between phases ($p < 0.001$). There were no statistically significant differences in the GRF impulse of the UW phase ($p = 0.472$), but the GRF impulse increased by 5.0% during the W phase ($p < 0.001$). Furthermore, the correlation ($r = -0.110$) between the increase in VJ height and increase in peak GRF during the W phase was not significant ($p > 0.05$).

Gender

Male athletes, on average, had a 48% higher VJ than did female athletes ($p < 0.001$), but gender $\times$ time interaction was not significant ($p = 0.594$). Similarly, male athletes had a 42% larger peak VGRF than did female athletes ($p < 0.001$), but the gender $\times$ time interaction was not significant ($p = 0.416$). Taken together, these results indicate that the changes in VJ height and peak VGRF were not affected by gender. Chi-square analyses revealed that neither a particular exercise ($\chi^2 = 0.311$; $p = 0.577$) nor a particular rest interval ($\chi^2 = 1.400$; $p = 0.237$) was more advantageous for men or women.

DISCUSSION

The purpose of this study was to determine, both individually and for the group mean, if (a) a power exercise led to greater PAP than a strength exercise; (b) a 4- or 5-minute rest interval led to greater PAP; and (c) if the effect of PAP was an increase in the peak GRF during a VJ. The influence of gender also was examined. Our findings do not support the hypothesis that a power exercise would be more effective at eliciting a PAP than a strength exercise. The 4-minute rest interval led to a

TABLE 4. Vertical ground reaction force characteristics between pre and postconditions.

Phase	Variable	Pre	Post
Unweighting	Total time (s)	0.44 (0.15)	0.42 (0.19)
	Peak force (N)*	-0.58 (0.27)	-0.68 (0.25)
	Impulse (N-s)	-0.18 (0.24)	-0.13 (0.04)
Weighting	Total time (s)*	0.50 (0.10)	0.47 (0.08)
	Peak force (N)	1.50 (0.19)	1.54 (0.20)
	Impulse (N-s)*	0.40 (0.05)	0.42 (0.05)
	Peak to peak difference (N)*	2.08 (0.25)	2.22 (0.24)

* $p < 0.05$.

significant increase in performance for group mean data, whereas the 5-minute rest interval did not. However, when examining each subject individually, we found that some subjects had a higher potentiation with the squat, whereas others had a higher potentiation with the clean. Likewise, for some subjects, the 5-minute rest interval was superior to the 4-minute rest interval in improving VJ height. Taken together, our results indicate that PAP is a highly individualized phenomenon but does not appear to be influenced by gender. Finally, the increased performance because of PAP was a result of an increase in the GRF impulse without necessarily increasing the peak GRF.

Few studies have investigated the use of a “power” exercise to elicit PAP. McBride et al. found that loaded countermovement jumps did not significantly potentiate 40-m sprint performance (14), but a heavy back squat did. In contrast, Gilbert and Lees found similar improvements in VJ height after PAP protocols involving a back squat with either a 1RM or the load that produced maximum peak power (7). Our results did not show a significant effect for exercise (squat vs. clean), indicating that both were equally effective in acutely increasing performance for the group mean.

Exercise produces both potentiation and fatigue effects. The fatigue effects are thought to predominate immediately post-exercise, but the potentiation effects last longer (19). Thus, increases in performance require a precise timeline in which the rest interval is long enough for the potentiation effects to be greater than the fatigue effects, but not so long that the potentiation effects decrease back to baseline. In trying to establish this timeline, rest periods have been investigated from 10 seconds (12) to over 15 minutes (3,13), with equivocal results.

The rest periods (4 and 5 minutes) were not necessarily chosen because they were thought to be the optimal rest periods to elicit PAP. Rather, a rest period (4 minutes) was chosen because it was among the shortest thought to elicit a PAP, but produced equivocal results in previous investigations (12,20). The other rest interval was chosen to determine literally how much difference a minute could make. For example, after the clean, 4 subjects (6,9,12,14) had a decrease in VJ height after 4 minutes of rest, but an increase after 5 minutes; conversely, 4 subjects (3,4,11,13) had an increase in VJ height after 4 minutes of rest, but a decrease after 5 minutes. Even with these (relatively) short timelines, every individual in our sample saw an increase in VJ height, but further investigations are required to determine the effect of longer or shorter rest periods.

Several authors have either reported (3) or suggested (4,5,17) that PAP is an individual phenomenon. Even with our homogeneous sample (all Division I volleyball players with no established trends between genders), our results indicate that the optimal exercise/rest interval to the optimal exercise and rest interval combination elicit PAP is highly individualized. The group mean increase in jump height of ~2% is similar to the value of others who have reported an increase in performance post-PAP (8,20). When the exercise/rest

interval exercise and rest interval combination that produced the highest increase in performance was examined, the group mean change in jump height increased to almost 6%.

On examining the single-subject results, we see that nearly one-third of our subjects had a greater potentiation effect with the hang clean, although the majority did see larger performance improvements with the back squat. Although a majority of subjects had their largest PAP with 4 minutes of rest, a quarter of them (25%) responded better to a 5-minute rest interval. The fact that the rest interval producing the largest performance is highly individualized combined with the finding that PAP after a power exercise may have a different optimum rest period compared with a strength exercise (7) may make finding a “one size fits all” exercise/rest period protocol exercise and rest interval combination for PAP effects tenuous at best. General programs may only impact a small percentage of athletes.

The GRF impulse (i.e., the integration of the GRF–time curve) determines the velocity at take-off, which in turn determines jump height (21). The GRF impulse can be increased by increasing the magnitude of the GRF, the duration of the GRF, or the overall shape of the GRF profile (by increasing the amount of time spent near the peak magnitude). Several investigations have found a small but insignificant increase in the peak GRF of the VJ after PAP (4,12,13). Our investigation supports the notion that PAP does not improve performance by increasing the peak GRF. Additionally, our results show a significant decrease in the time of the W phase with no increase in the peak magnitude of the GRF. Thus, improvements in performance of the VJ after PAP are related to the changes in the overall shape of the GRF profile, which is in agreement with previous findings that PAP protocols lead to increases in isometric rate of force development, but not to increases in isometric maximum force (7) and that the overall shape of the GRF profile is more important than the peak magnitude in determining jump height (6).

With the techniques used in our investigation, the increase in the negative peak during the UW phase defies precise mechanical explanation. It does not represent the entire eccentric phase of the countermovement. Rather, it is proportional to the peak acceleration in the downward direction before take-off. A larger negative acceleration could lead to more rapid and longer stretching of the prime movers (hip extensors, knee extensors, and plantar flexors), which could enhance force production concentrically through both elastic recoil of the elastic components and “preforcing” of the muscle before the concentric muscle action (21). However, more research, using a variety of sophisticated biomechanical analyses, is necessary before coming to a definitive conclusion or explaining why this would be a characteristic of PAP.

In summary, our results indicate that even in a homogenous sample, PAP is highly individualized, and the force–time characteristics of the GRF are altered without necessarily increasing the peak force. Our results do not imply that either 4 or 5 minutes of rest is the optimal rest interval to elicit PAP.

Some of our subjects may have had larger improvements in VJ height with a shorter rest period, and others with a longer one. More research is necessary to determine the optimal exercise/rest interval exercise and rest interval combination for an individual without having to resort to trial and error.

PRACTICAL APPLICATIONS

Complex training can produce significant, acute increases in VJ height if the proper exercise and rest interval is chosen for each individual. Unfortunately, at present, trial and error must be used to determine which athletes would respond better to a particular exercise or rest interval. Future research should focus on the individual effects of determining intracomplex rest intervals and the heavy loaded resistance exercise that precedes the plyometric exercise, and the long-term effects complex training has on jump performance.

REFERENCES

1. Bates, BT. Single subject methodology: An alternative approach. *Med Sci Sports Exerc* 28: 631–638, 1996.
2. Chiu, LZ and Barnes, JL. The fitness-fatigue model revisited: Implications for planning short- and long-term training. *Strength Cond J* 25: 42–51, 2003.
3. Chiu, LZ, Fry, AC, Weiss, LW, Schilling, BK, Brown, LE, and Smith, SL. Postactivation potentiation response in athletic and recreationally trained individuals. *J Strength Cond Res* 17: 671–677, 2003.
4. Comyns, TM, Harrison, AJ, Hennessy, LK, and Jensen, RL. The optimal complex training rest interval for athletes from anaerobic sports. *J Strength Cond Res* 20: 471–476, 2006.
5. Duthie, GM, Young, WB, and Aitken, DA. The acute effects of heavy loads on jump squat performance: An evaluation of the complex and contrast methods of power development. *J Strength Cond Res* 16: 530–538, 2002.
6. Garhammer, J and Gregor, R. Propulsive forces as a function of intensity for weightlifting and vertical jumping. *J Appl Sport Sci Res* 6: 129–134, 1992.
7. Gilbert, G and Lees, A. Changes in the force development characteristics of muscle following repeated maximum force and power exercise. *Ergonomics* 48: 1576–1584, 2005.
8. Gourgoulis, V, Aggeloussis, N, Kasimatis, P, Mavromatis, G, and Garas, A. Effect of a submaximal half-squats warm-up program on vertical jumping ability. *J Strength Cond Res* 17: 342–344, 2003.
9. Hamada, T, Sale, DG, MacDougall, JD, and Tarnopolsky, MA. Postactivation potentiation, fiber type, and twitch contraction time in human knee extensor muscles. *J Appl Physiol* 88: 2131–2137, 2000.
10. Hill, AV. The heat of shortening and the dynamic constants of muscle. *Proc Roy Soc Lond B: Biol Sci* 126: 136–195, 1938.
11. Hodgson, M, Docherty, D, and Robbins, D. Post-activation potentiation—Underlying physiology and implications for motor performance. *Sports Med* 35: 585–595, 2005.
12. Jensen, RL and Ebben, WP. Kinetic analysis of complex training rest interval effect on vertical jump performance. *J Strength Cond Res* 17: 345–349, 2003.
13. Jones, P and Lees, A. A biomechanical analysis of the acute effects of complex training using lower limb exercises. *J Strength Cond Res* 17: 694–700, 2003.
14. McBride, JM, Nimphius, S, and Erickson, TM. The acute effects of heavy-load squats and loaded countermovement jumps on sprint performance. *J Strength Cond Res* 19: 893–897, 2005.
15. Miyamoto, N, Fukunaga, T, and Kawakami, Y. Evidence for intermuscle difference in postactivation potentiation in the human triceps surae: a mechanomyographic study. *Muscle Nerve* 39: 703–706, 2009.
16. Rixon, KP, Lamont, HS, and Bembien, MG. Influence of type of muscle contraction, gender, and lifting experience on postactivation potentiation performance. *J Strength Cond Res* 21: 500–505, 2007.
17. Scott, SL and Docherty, D. Acute effects of heavy preloading on vertical and horizontal jump performance. *J Strength Cond Res* 18: 201–205, 2004.
18. Stone, MH and O'Bryant, HS. *Weight Training: A Scientific Approach*. Edina, MN: Burgess International Group, Inc. 1987.
19. Tillin, NA and Bishop, D. Factors modulating post-activation potentiation and its effect on performance of subsequent explosive activities. *Sports Med* 39: 147–166, 2009.
20. Young, WB, Jenner, A, and Griffiths, K. Acute enhancement of power performance from heavy load squats. *J Strength Cond Res* 12: 82–84, 1998.
21. Zajac, FE. Muscle coordination of movement—A perspective. *J Biomech* 26: 109–124, 1993.